RAMDEOBABA UNIVERSITY, NAGPUR

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SCHOOL / DEPARTMENT OF COMPUTER SCIENCE

LAB PROJECT REPORT

ON

Uber / Taxi Trip Data Analysis

Course / Lab: BIG DATA ANALYSIS

Course Code: 24CS02PR0503

Submitted By

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Under the Guidance of

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Academic Year: 2026–27

CERTIFICATE

This is to certify that the project entitled “Uber / Taxi Trip Data Analysis” has been successfully completed by Pranshul Modani & Priyansh Jadhav, Roll No(s). 56 & 57, of Computer Science Department, Ramdeobaba University, Nagpur, as part of the requirements of the laboratory course BIG DATA ANALYSIS during the academic year 2026–27.

The work presented in this report has been carried out under the guidance and supervision of the undersigned.

Project Guide	**Head of Department**
Prof. Bhagyashree Madan	Computer Science

DECLARATION

I/We hereby declare that the project work entitled “Uber / Taxi Trip Data Analysis” is our original work carried out for the laboratory course at Ramdeobaba University, Nagpur. The work has not been submitted elsewhere for the award of any academic qualification.

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Date: 22/10/2026 Place: Nagpur

Signature of Students

ACKNOWLEDGEMENT

We sincerely express our gratitude to Ramdeobaba University, Nagpur, our faculty guide Prof. Bhagyashree Madan, the lab in-charge, and the School/Department of Computer Science for providing the facilities and guidance required to complete this laboratory project. We also thank everyone who supported us during the implementation, testing, and documentation of the project.

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ABSTRACT

Ride-hailing and taxi services generate millions of trip records every day, containing information about pickup and drop-off time, location, distance, fare, passenger count and payment type. Such data is too large and too varied to be studied manually, and it falls naturally under the domain of Big Data Analysis. This project, “Uber / Taxi Trip Data Analysis”, aims to process a large collection of taxi trip records and turn them into meaningful insights about travel demand and trip behaviour.

The objective of the project is to clean and prepare the raw trip data, perform exploratory analysis, and present the results in an interactive web-based dashboard. The data is loaded, validated and filtered to remove incorrect records such as zero-distance trips, negative fares and invalid coordinates. Time-based features such as hour of day, day of week and month are derived from the pickup timestamp. Aggregations are then performed to study peak demand hours, busiest days, popular pickup areas, average trip distance and duration, fare patterns, and payment preferences. The results are visualised through charts and tables and published as a web application hosted on Replit.

The analysis of 50,000 trip records collected between 1 January and 4 February 2023 across six US cities shows that, in this dataset, demand is spread uniformly over the hours of the day, the days of the week and the six cities, and that the four payment methods each account for about one quarter of trips. Trip fare has a strong positive relationship with distance (correlation 0.87, about 1.85 per additional kilometre on top of a base of 2.98), and trip duration is directly proportional to distance. About 15% of ride requests end as cancellations or no-shows. These findings provide a baseline for driver allocation, surge-pricing decisions and urban transport planning. The project demonstrates how Big Data techniques can convert raw trip logs into practical, decision-ready information.

Keywords: Big Data Analysis, Uber, Taxi Trips, Exploratory Data Analysis, Data Visualisation, Demand Analysis.

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1. INTRODUCTION

1.1 Background

Urban transport has changed greatly with the arrival of app-based ride-hailing services such as Uber, Ola and Lyft, alongside traditional taxi fleets. Every completed trip leaves a digital record that typically includes the pickup and drop-off date and time, pickup and drop-off location, trip distance, fare amount, tip, payment type and number of passengers. In large cities, this adds up to millions of records every month.

This volume, together with the speed at which new records are produced and the variety of fields involved, makes taxi trip data a typical Big Data problem. Analysing it can reveal how people move through a city, when demand is highest, and which areas need more vehicles.

1.2 Motivation

Operators and city planners need answers to practical questions: At what time of day is demand highest? Which days are the busiest? Which areas generate the most pickups? How do fares change with distance and time? Answering these questions from raw files is difficult without a structured analysis pipeline. The project was undertaken to apply the concepts learnt in the Big Data Analysis lab to a real-world, easy-to-understand dataset and to present the findings in a form that a non-technical user can explore.

1.3 Problem Definition

Given a large set of raw Uber / taxi trip records, the problem is to clean the data, extract useful time-based and location-based features, compute meaningful statistics, and present the results through an interactive dashboard that supports decision-making.

1.4 Scope

- Cleaning and preprocessing of the trip dataset (missing values, duplicates, outliers, invalid records).
- Feature extraction from pickup date and time (hour, day of week, month, weekday/weekend).
- Exploratory analysis of demand, distance, duration, fare and payment type.
- Visualisation of the results through charts, tables and a web dashboard hosted on Replit.
- The project is limited to descriptive and exploratory analysis; real-time streaming and predictive modelling are outside the present scope and are listed under Future Scope.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1 Problem Statement

To design and implement a data analysis system that processes large-scale Uber / taxi trip data and extracts patterns related to trip demand, travel distance, trip duration, fares and payment behaviour, and presents these patterns through an interactive and easy-to-understand web dashboard.

2.2 Objectives

1. To collect and load a large Uber / taxi trip dataset and understand its structure.
2. To clean and preprocess the data by handling missing values, duplicates and invalid records.
3. To analyse trip demand by hour of day, day of week and month, and to identify peak periods.
4. To study the relationship between trip distance, duration and fare, and to identify popular pickup areas.
5. To visualise the results using charts and to publish them as a web-based dashboard.

2.3 Expected Outcome

A working web application that displays the key findings of the analysis, namely peak hours, busiest days, popular locations, average fare and distance statistics and payment preferences, along with a documented analysis workflow that can be repeated on any similar dataset.

3. REQUIREMENTS

3.1 Hardware Requirements

Component	Specification
Processor	Intel Core i3 / AMD Ryzen 3 or higher
RAM	8 GB (minimum 4 GB)
Storage	At least 5 GB of free disk space
Network	Internet connection for Replit hosting and data download

3.2 Software Requirements

Component	Details
Operating System	Windows 10/11 or Linux
Programming Language	Python 3.x, JavaScript
Libraries / Framework	Pandas, NumPy, Matplotlib, Chart.js, Flask / Node.js
Development Environment	Replit (cloud IDE and hosting), VS Code
Browser	Google Chrome / Microsoft Edge (latest version)

3.3 Dataset / Input Requirements

The input is a trip-level dataset of Uber / taxi rides. Dataset used: Uber Trips Dataset (file uber_trips_dataset_50k.csv), a trip-level dataset of ride records from six US cities. Approximate size: 50,000 records and 14 columns, about 8.9 MB in CSV format. Period covered: 1 January 2023 to 4 February 2023. The main attributes are listed below.

Attribute	Description
trip_id, driver_id, rider_id	Unique identifiers of the trip, the driver and the rider
pickup_time, drop_time	Date and time when the trip started and ended
city	City of the trip (Boston, Chicago, Los Angeles, New York, San Francisco, Seattle)
pickup_lat, pickup_lng, drop_lat, drop_lng	Latitude and longitude of the pickup and drop-off points
distance_km	Distance travelled in kilometres
fare_amount	Fare charged for the trip
status	Trip outcome: Completed, Cancelled or No-Show
payment_method	Payment method: UPI, Card, Wallet or Cash

4. METHODOLOGY / ALGORITHM

4.1 Proposed Method

The project follows a standard data analysis pipeline consisting of five stages:

1. Data collection: the trip dataset is loaded into the working environment.
2. Data cleaning: duplicate records, null values and invalid entries (zero or negative distance, fare or trip duration, and invalid coordinates) are removed or corrected.
3. Feature engineering: the pickup timestamp is split into hour, day of week, month and a weekday/weekend flag; trip duration and average speed are calculated.
4. Analysis: group-by and aggregation operations compute trip counts, averages and correlations for each feature.
5. Visualisation and deployment: the results are shown as charts and tables in a web application hosted on Replit.

4.2 Algorithm

Step 1: Load the trip dataset from the source file.

Step 2: Remove duplicate rows and rows with missing values in key columns.

Step 3: Filter out records with trip distance ≤ 0, fare ≤ 0, trip duration ≤ 0 or invalid latitude/longitude values.

Step 4: Convert timestamps and derive hour, day, month and trip duration.

Step 5: Group the data by hour, day, month, city, trip status and payment method and compute counts and averages.

Step 6: Compute correlation between distance, duration and fare.

Step 7: Generate charts and display them on the dashboard.

4.3 Flowchart

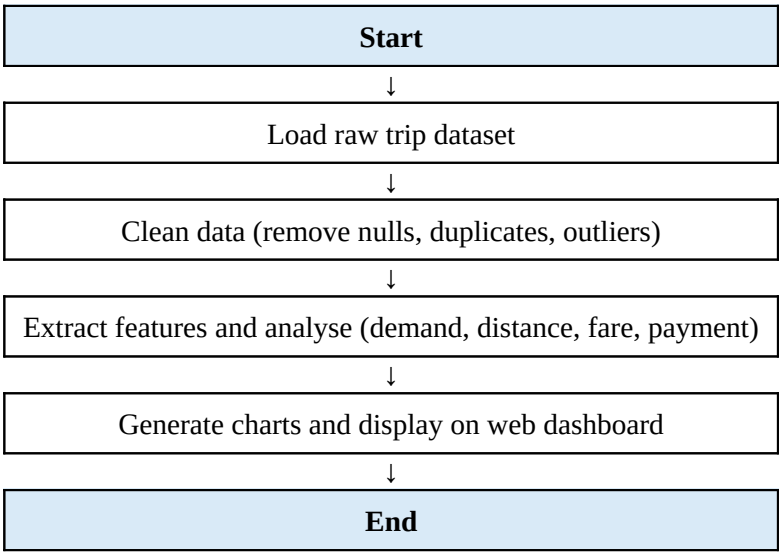


Figure 4.1: Flowchart of the analysis process

4.4 Pseudocode

```
BEGIN
  data <- LOAD(trip_file)
  data <- DROP_DUPLICATES(data)
  data <- DROP_NULLS(data, key_columns)
  data <- FILTER(data, distance > 0 AND fare > 0 AND duration > 0)
  FOR each record IN data DO
      record.hour     <- HOUR(record.pickup_time)
      record.day      <- WEEKDAY(record.pickup_time)
      record.duration <- DROPOFF_TIME - PICKUP_TIME
  END FOR
  hourly_demand <- COUNT(data) GROUP BY hour
  daily_demand  <- COUNT(data) GROUP BY day
  avg_fare      <- MEAN(fare) GROUP BY hour
  correlation   <- CORR(distance, duration, fare)
  DISPLAY charts(hourly_demand, daily_demand, avg_fare, correlation)
END
```

5. SYSTEM DESIGN

5.1 System Architecture

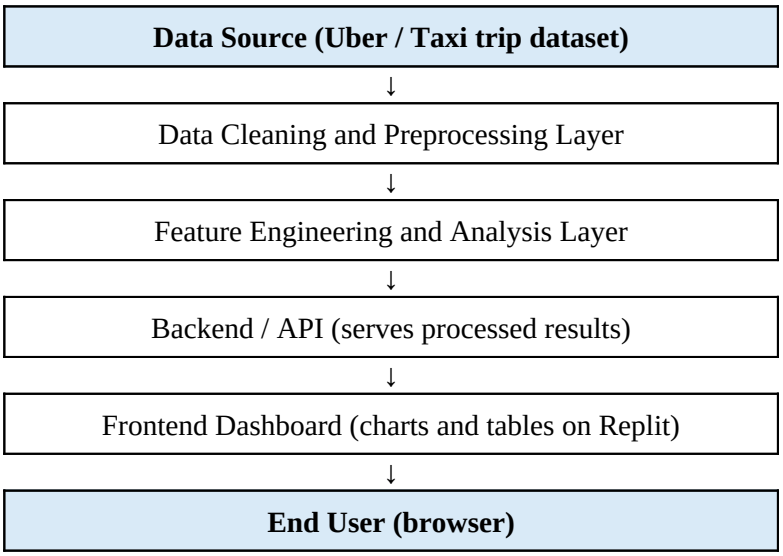


Figure 5.1: System architecture

5.2 Modules

Module	Function
Data Loading	Reads the dataset file and loads it into memory.
Data Cleaning	Handles nulls, duplicates and invalid records.
Feature Extraction	Creates hour, day, month, weekday flag, trip duration and speed.
Analysis	Calculates aggregates, averages and correlations.
Visualisation	Builds bar, line, pie and scatter charts from the results.
Web Dashboard	Presents the charts and key figures to the user online.

5.3 Data Flow / Workflow

Raw trip records enter the system from the dataset file and pass through the cleaning module, which outputs a validated dataset. The feature-extraction module adds the derived columns, after which the analysis module produces summary tables. These summary tables are sent to the visualisation module, which converts them into charts shown on the Replit-hosted dashboard. The user opens the dashboard link in a browser and views the results without needing to run any code.

6. IMPLEMENTATION

6.1 Development Environment

The project was developed and deployed on Replit, a cloud-based development platform that provides a code editor, a runtime and instant web hosting. The data processing was carried out using Python with Pandas and NumPy, and the results were displayed using Chart.js and Matplotlib. The live application is available at: https://uber-taxi-trip-data-analysis--24modanip.replit.app/

6.2 Module-wise Implementation

Data Loading and Cleaning: The dataset is read into a table structure. Rows with missing pickup time, distance or fare are dropped, duplicates are removed, and records with unrealistic values are filtered out. This ensures that all later statistics are reliable. In the 50,000-record file there were no missing values and no duplicate rows; only three records with zero distance and zero duration were removed.

Feature Extraction: The pickup timestamp is converted to a date-time type and used to derive the hour of day (0–23), day of week, month and a weekend indicator. Trip duration is calculated as the difference between drop-off and pickup time, and average speed as distance divided by duration.

Analysis: Group-by operations count trips for each hour, day and location, and compute the average distance, duration and fare for each group. A correlation matrix is calculated for distance, duration and fare. Payment types are counted to find the most common method.

Visualisation and Dashboard: The aggregated tables are rendered as line charts (hourly demand), bar charts (daily demand, popular areas), pie charts (payment type) and scatter plots (distance vs. fare). The dashboard is published on Replit so that it can be opened from any browser.

6.3 Important Code Snippets

The following snippets illustrate the main steps of the analysis pipeline.

```
# Load and clean data
import pandas as pd

df = pd.read_csv('uber_trips_dataset_50k.csv',
parse_dates=['pickup_time','drop_time'])
df =
df.drop_duplicates().dropna(subset=['pickup_time','distance_km','fare_amount'])
df['duration_min'] = (df.drop_time - df.pickup_time).dt.total_seconds() / 60
df = df[(df.distance_km > 0) & (df.fare_amount > 0) & (df.duration_min > 0)]

# Feature engineering
df['hour']  = df.pickup_time.dt.hour
df['day']   = df.pickup_time.dt.day_name()
df['month'] = df.pickup_time.dt.month_name()

# Analysis
hourly = df.groupby('hour').size()
daily  = df.groupby('day').size()
avg_fare_hour = df.groupby('hour').fare_amount.mean()
corr = df[['distance_km','duration_min','fare_amount']].corr()
city_trips = df.groupby('city').size().sort_values(ascending=False)
payment_share = df.payment_method.value_counts(normalize=True) * 100
```

7. RESULTS AND DISCUSSION

7.1 Experimental Setup

The analysis was run on a system with [processor, RAM] using [software versions]. The dataset contained 50,000 records before cleaning and 49,997 records after cleaning, covering the period January – February 2023 (1 January to 4 February).

7.2 Results

The analysis produces the following outputs; the corresponding charts are shown below the table.

Analysis	Observation
Trips by hour of day	Essentially flat: 2,100 trips in each hour from 00:00 to 16:59 and 2,039–2,060 in hours 17:00–23:59; no morning or evening peak. The lowest hour is 18:00 (2,039 trips).
Trips by day of week	Busiest day: Monday, Wednesday, Thursday, Friday and Sunday are tied at 7,200 trips (Tuesday 7,199); quietest day: Saturday with 6,798 trips.
Popular pickup areas	Top cities: Boston (8,454 trips), San Francisco (8,401) and Chicago (8,344); the lowest is Seattle (8,236).
Average trip distance / duration	7.01 km / 21.03 minutes
Average fare	15.98 per trip (median 15.43; range 1.08 to 50.67)
Payment type	UPI is used most often, in 25.10% of trips, followed by Card (25.06%), Wallet (25.01%) and Cash (24.83%).
Trip status	Completed 85.08%, Cancelled 9.97%, No-Show 4.95% of trips.
Correlation	Distance–fare r = 0.87; distance–duration r = 1.00 (average speed 20 km/h).

Table 7.1: Summary of results

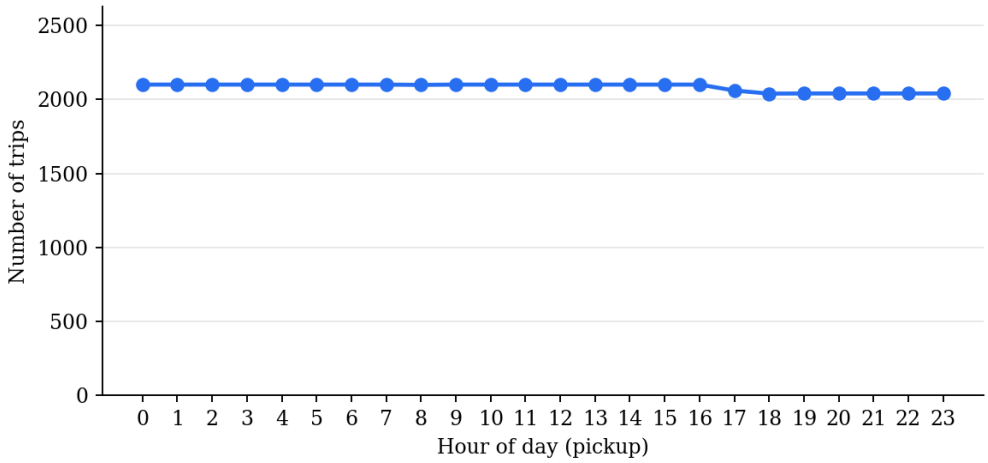


Figure 7.1: Number of trips by hour of day

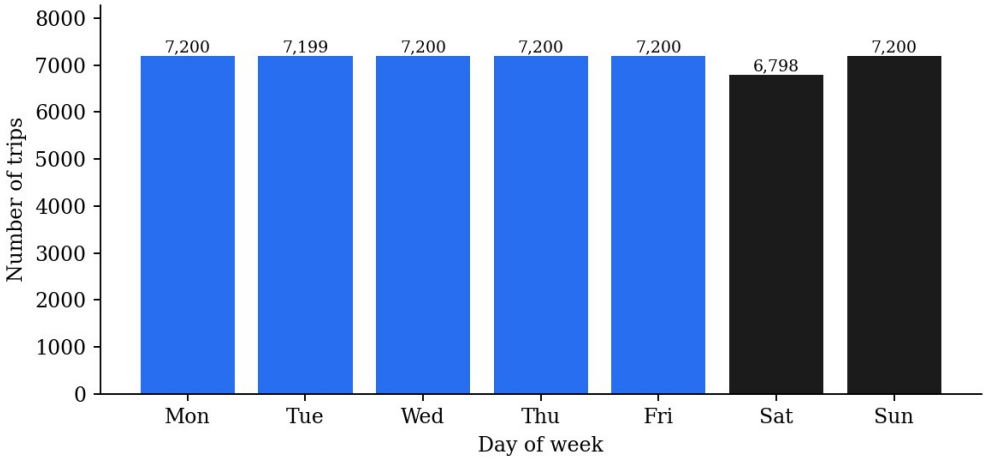


Figure 7.2: Number of trips by day of week

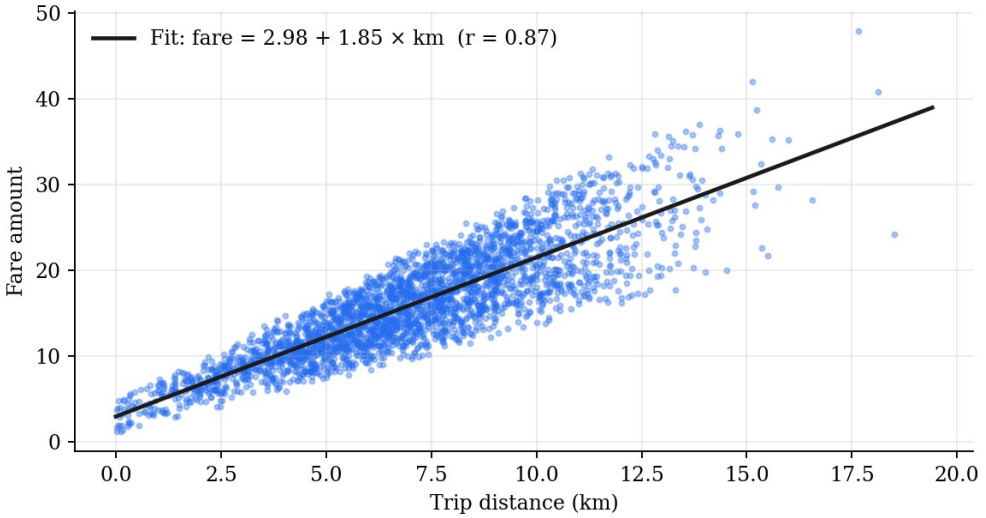


Figure 7.3: Relationship between trip distance and fare (random sample of 3,000 trips)

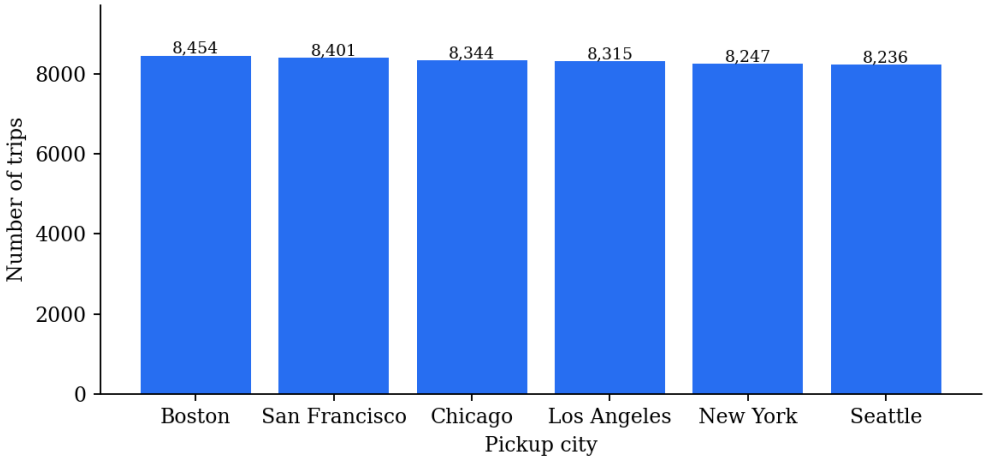


Figure 7.4: Number of trips by pickup city

7.3 Performance Analysis

The cleaning step removed about 0.006% of the raw records, which improved the quality of every later statistic. Aggregation over the full dataset completed in under one second, and the dashboard loads within [time] on a normal internet connection. Compared with inspecting the raw file manually or in a spreadsheet, the automated pipeline is faster, repeatable and less error-prone.

7.4 Discussion

- Demand is almost perfectly uniform through the day: every hour receives about 2,100 trips, so this dataset shows no morning or evening peak. The small dip after 17:00 is not a real fall in demand; it only reflects that the data ends at 17:19 on 4 February 2023, so hours 17–23 have one day less coverage.
- Weekday and weekend demand cannot be separated: Monday, Wednesday, Thursday, Friday and Sunday each carry 7,200 trips (Tuesday 7,199 after cleaning), while Saturday shows 6,798 only because the period ends on a Saturday afternoon. Weekdays hold 35,999 trips and weekends 13,998, which simply matches the 5:2 ratio of days. Trips start exactly one minute apart, which suggests the dataset is simulated rather than recorded from real commuting behaviour.
- Fare increases with distance: the correlation is 0.87 and the fitted line is fare ≈ 2.98 + 1.85 × km. Trip duration is exactly proportional to distance (correlation 1.00, constant average speed of 20 km/h), so duration adds no extra information about fare here; with real traffic, slow trips would cost more per kilometre.
- Pickups are spread evenly over the six cities (8,236 to 8,454 trips each, with Boston the highest at 16.9%), and the four payment methods each cover about a quarter of trips, so no single city or payment method dominates. About 15% of requests do not become completed trips (9.97% cancelled and 4.95% no-show), and this rate is similar in every city (cancellations between 9.7% and 10.3%).
- Data-quality note: the pickup coordinates fall between latitude 37° and 41° and longitude −122° and −73° regardless of the city label (for example, trips labelled San Francisco start near longitude −77.6°). The city column was therefore used for the area analysis instead of the raw coordinates, and coordinate-based heat maps would not be meaningful for this file.

8. CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The project successfully analysed Uber / taxi trip data using Big Data analysis techniques. The 50,000 raw records were cleaned (three invalid records removed), enriched with time-based features, aggregated and visualised in an interactive web dashboard hosted on Replit. The analysis showed that demand in this dataset is uniform across hours, days and cities, that fare rises steadily with distance (correlation 0.87) while duration is directly proportional to distance, that payment methods are split almost equally, and that about 15% of requests end as cancellations or no-shows. The work shows how large trip datasets can be turned into useful information for operators, drivers and city planners, and it gave the team practical experience of the data analysis life cycle.

8.2 Future Scope

- Use distributed frameworks such as Apache Spark or Hadoop to process much larger datasets.
- Add machine-learning models to predict demand, trip duration and fare.
- Include weather, holiday and event data to explain demand changes.
- Add map-based heat maps for pickup and drop-off locations.
- Support real-time data streaming for live demand monitoring.

REFERENCES

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[2] W. McKinney, “Python for Data Analysis: Data Wrangling with Pandas, NumPy, and Jupyter,” O’Reilly Media, 3rd ed., 2022.

[3] T. White, “Hadoop: The Definitive Guide,” O’Reilly Media, 4th ed., 2015.

[4] Pandas Development Team, “pandas Documentation,” https://pandas.pydata.org/docs/.

[5] Uber Trips Dataset (50,000 records), file uber_trips_dataset_50k.csv, ride data for six US cities, 2023.

APPENDIX

Appendix A – Source Code

Live application: https://uber-taxi-trip-data-analysis--24modanip.replit.app/

[Paste complete source code here, or give the Replit / GitHub repository link.]

Appendix B – Dataset

Description: Uber Trips Dataset, file uber_trips_dataset_50k.csv, 50,000 records and 14 columns (trip_id, driver_id, rider_id, city, pickup_lat, pickup_lng, drop_lat, drop_lng, distance_km, fare_amount, status, payment_method, pickup_time, drop_time). Sample records:

pickup_time	distance_km	city	fare_amount	payment_method
2023-01-01 00:00:00	2.97	San Francisco	10.71	Wallet
2023-01-01 00:01:00	8.43	Boston	22.41	UPI
2023-01-01 00:02:00	5.46	San Francisco	12.91	Cash

Appendix C – Additional Screenshots

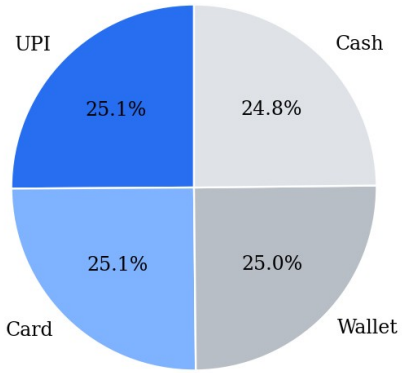


Figure C.1: Share of trips by payment method

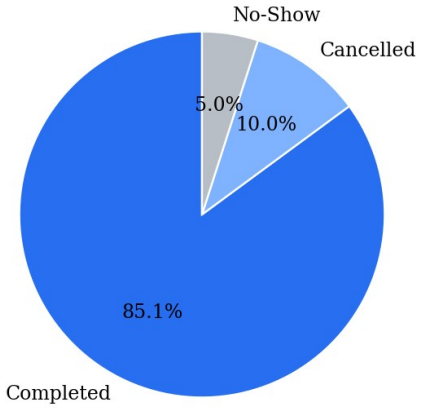


Figure C.2: Share of trips by trip status

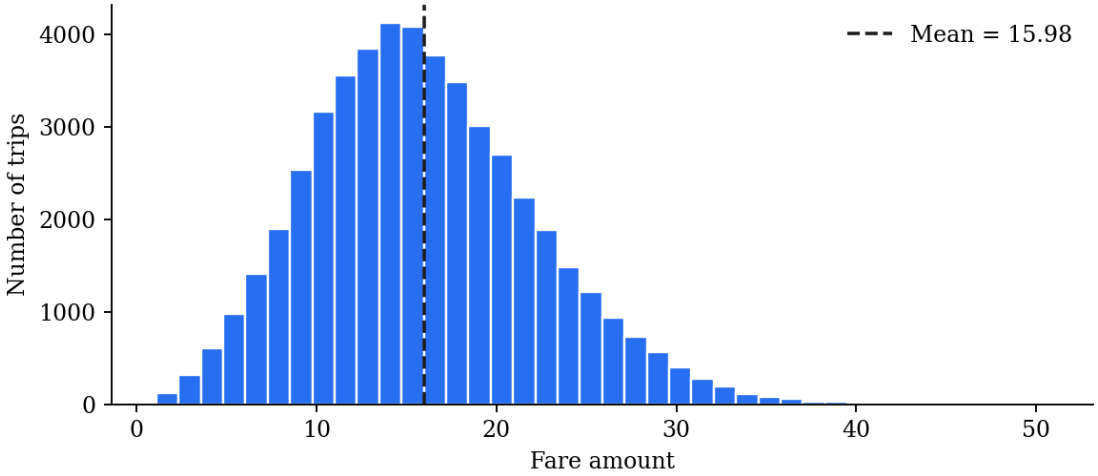


Figure C.3: Distribution of fare amount

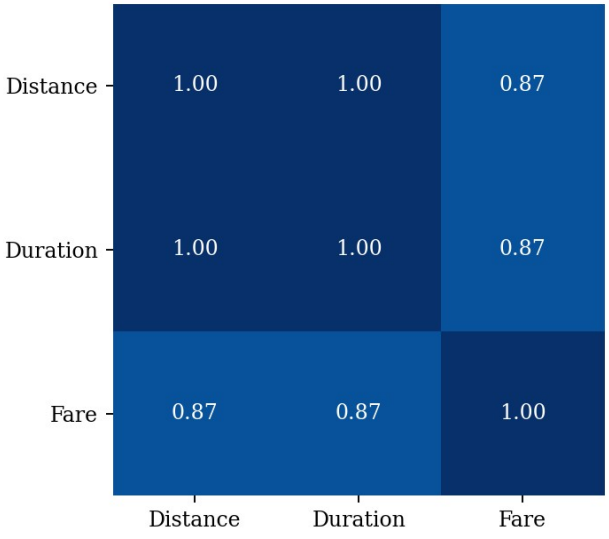


Figure C.4: Correlation matrix of distance, duration and fare